

Agentic AI-Powered Smart Campus Grievance Management System

Dr. J. Anitha

*Dept. of Artificial Intelligence
and Data Science*

*Sri Ramakrishna Engineering College
Coimbatore, India
anitha.j@srec.ac.in*

Akshayalakshmi S

*Dept. of Artificial Intelligence
and Data Science*

*Sri Ramakrishna Engineering College
Coimbatore, India
akshayalakshmi.2311@srec.ac.in*

Nikitha M

*Dept. of Artificial Intelligence
and Data Science*

*Sri Ramakrishna Engineering College
Coimbatore, India
nikitha.2311@srec.ac.in*

Abstract—Traditional campus grievance handling systems rely heavily on manual registration, classification, and routing of complaints, leading to delayed responses, poor prioritization, and weak inter-departmental coordination. This paper presents an Agentic AI-Powered Smart Campus Grievance Management System that autonomously perceives, classifies, and resolves infrastructure-related complaints using computer vision and a multi-agent architecture. A Vision Agent powered by MobileNetV2 with transfer learning classifies uploaded complaint images into six infrastructure categories with an overall macro F1-score of 91.4%. A Classification and Decision Agent maps predicted categories to the responsible campus department and assigns urgency levels using a rule-based priority function. A Notification Agent automates email communication to administrators and users, while a Monitoring Agent tracks each complaint through a three-state lifecycle in real time. The system is implemented using FastAPI and Python with a web-based administrator dashboard. Evaluation demonstrates a 63% reduction in average complaint processing time and a 62.5% reduction in resolution time compared to manual workflows. A two-week pilot deployment with 87 real student complaints achieved 100% automated processing with a user satisfaction score of 4.3 out of 5.0.

Index Terms—Agentic AI, Smart Campus, Grievance Management System, Multi-Agent System, Computer Vision, MobileNetV2, Transfer Learning, Autonomous Decision Making

I. INTRODUCTION

Educational campuses are complex operational environments that continuously generate infrastructure-related issues including road damage, water leakage, malfunctioning lights, garbage accumulation, door damage, and general building maintenance problems. Effective resolution of these issues requires rapid detection, accurate classification, appropriate department routing, and timely follow-up. In most institutions, however, grievance resolution is carried out manually or in a semi-automated manner, placing a significant administrative burden on staff and resulting in delayed responses, inconsistent prioritization, and inadequate monitoring of complaint status.

Traditional campus grievance management systems primarily serve as record-keeping tools, offering basic complaint registration and status lookup functionality. They lack the capacity to intelligently interpret visual data submitted by users, automate department assignment, or adapt dynamically

to varying complaint loads. Existing approaches rely on rule-based categorization with no support for image understanding or real-time analytics [1], [2]. Campus helpdesk and ERP systems provide complaint tracking but require manual staff verification before forwarding, introducing significant delays [4].

Recent advances in Agentic Artificial Intelligence (AI) and multi-agent systems offer a compelling framework for overcoming these limitations. Agentic AI refers to systems composed of autonomous agents capable of perceiving the environment, reasoning about situations, executing actions, and monitoring outcomes without continuous human guidance [7]. When applied to grievance management, individual agents can handle distinct responsibilities—perception, decision-making, notification, and monitoring—and coordinate seamlessly to deliver end-to-end automation.

In this context, this paper proposes an Agentic AI-Powered Smart Campus Grievance Management System in which image-based grievance handling is fully automated using four specialized intelligent agents: (i) a Vision Agent for automated issue detection and image classification, (ii) a Classification and Decision Agent for department assignment and urgency prediction, (iii) a Notification Agent for automated email communication, and (iv) a Monitoring Agent for real-time complaint lifecycle tracking. The primary objectives are to enhance efficiency, accuracy, and transparency in campus grievance handling while minimizing reliance on manual intervention.

The remainder of this paper is organized as follows. Section II presents related work. Section III describes system design and agent architecture. Section IV details the mathematical models. Section V covers implementation. Section VI presents results and analysis. Section VII concludes with future directions.

II. RELATED WORK

Izadi [1] proposed CatIss, a transformer-based tool for automatic categorization of GitHub issue reports using natural language processing. While effective for text-based issues, the approach does not address visual complaint data or autonomous routing—limitations overcome in the proposed system through image-based input and agentic decision-making.

Ebrahim and Joy [2] explored few-shot issue report classification using adapter modules on pre-trained language models, demonstrating strong performance with limited labeled data. This is relevant to the low-data transfer learning strategy in the Vision Agent, though it focuses exclusively on textual software engineering artifacts.

Graeff [3] introduced a location-based crowdsourcing platform for civic engagement, motivating citizen-driven issue reporting. This work inspired the image-based campus complaint submission interface in the proposed system, extending the concept from public civic spaces to educational campuses.

Walwadkar et al. [4] presented a smart civic issue reporting system using mobile applications and SMS-based alerts. While providing basic complaint registration, it lacks AI-based classification, autonomous routing, or real-time monitoring—all central contributions of the proposed system.

Zangl et al. [5] analyzed transparent AI-driven toxicity detection tools for civic engagement platforms, emphasizing explainability and accountability. The monitoring agent and color-coded status indicators in the proposed system reflect similar transparency objectives.

Shah et al. [6] proposed recognizing public grievances from microblogs using NLP. This approach handles text-based social media inputs but does not support visual data or agent-based autonomous decision-making, distinguishing it from the proposed work.

In contrast to all prior work, the proposed system uniquely integrates image-based computer vision, a four-agent autonomous architecture, automated communication, and real-time monitoring into a unified campus-specific grievance management platform.

III. SYSTEM DESIGN AND METHODOLOGY

A. System Architecture

The proposed system follows a distributed multi-agent architecture in which each agent is responsible for a well-defined task in the grievance handling pipeline. Agents communicate through a centralized SQLite database and message-based internal API calls, ensuring loose coupling and modularity. The architecture consists of five core components: (i) a web-based complaint submission interface, (ii) intelligent image analysis via the Vision Agent, (iii) automated decision-making via the Classification and Decision Agent, (iv) communication automation via the Notification Agent, and (v) real-time status tracking via the Monitoring Agent. Fig. 1 illustrates the overall agent interaction and data flow.

The system operates as a layered control architecture. The *perception layer* handles raw image input and feature extraction. The *reasoning layer* performs classification and department assignment. The *action layer* executes notifications and dashboard updates. The *monitoring layer* continuously tracks complaint lifecycle states. This separation of concerns ensures that each agent can be independently updated or extended without affecting the rest of the system.

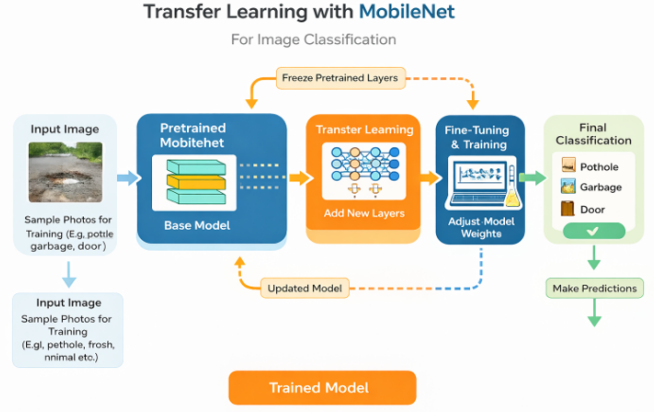


Fig. 1. Overall system architecture showing agent interactions and data flow.

B. Agent Modules

Table I summarizes the four agent modules and their responsibilities.

TABLE I
AGENT MODULES AND THEIR RESPONSIBILITIES

Agent	Responsibility
Vision Agent	Preprocesses uploaded image; extracts features via MobileNetV2; classifies into one of six categories: road damage, lighting, garbage, water leakage, door damage, or general campus issue
Classification & Decision Agent	Maps predicted class to responsible department via dept_map; assigns urgency (High/Medium/Low) using severity thresholding; generates unique complaint ID
Notification Agent	Dispatches automated email alerts to administrator and department head; sends acknowledgement receipt with complaint ID to user within 30 seconds
Monitoring Agent	Polls database every 30 s; detects state transitions (Not Seen → Noted → Resolved); updates color-coded dashboard indicators in real time

C. Workflow Control Logic

The system operates through four coordinated workflows triggered sequentially as part of the grievance processing pipeline.

Perception Workflow: Activated when a user submits an image through the web portal. The Vision Agent receives, preprocesses, and classifies the image, returning a predicted category label with a confidence score.

Decision Workflow: Triggered immediately after classification. The Classification and Decision Agent maps the predicted label to a campus department and computes an urgency score using the priority function defined in Section IV.

Communication Workflow: Executes after department assignment. The Notification Agent dispatches structured email alerts to the administrator and responsible department, and a confirmation receipt to the user, all within 30 seconds.

Monitoring Workflow: Runs continuously as a background process. The Monitoring Agent polls the complaint database,

detects administrator-triggered state transitions, and refreshes the dashboard status indicators accordingly.

IV. MODELS AND AGENT LOGIC

A. Image Classification Model – Vision Agent

Let the set of complaint images be represented as $\mathcal{I} = \{i_1, i_2, \dots, i_n\}$, where each image i_k is associated with a ground-truth label $y_k \in \mathcal{C} = \{\text{road damage, lighting, garbage, water leakage, door damage, general}\}$. Each image is resized to 224×224 pixels and normalized to $[0, 1]$ before processing. The MobileNetV2 backbone, pre-trained on ImageNet, produces a 1280-dimensional feature vector $\mathbf{f}_k$. A global average pooling layer followed by a fully-connected softmax classifier produces the predicted probability distribution:

$$\hat{y}_k = \text{softmax}(\mathbf{W} \cdot \mathbf{f}_k + \mathbf{b}) \quad (1)$$

where $\mathbf{W}$ and $\mathbf{b}$ are learnable parameters of the classification head. The predicted class is:

$$\hat{c}_k = \arg \max_{c \in \mathcal{C}} \hat{y}_k(c) \quad (2)$$

To address class imbalance, a class-weighted cross-entropy loss is used during fine-tuning:

$$\mathcal{L} = - \sum_k w_{y_k} \cdot \log(\hat{y}_k(y_k)) \quad (3)$$

where w_{y_k} is the inverse-frequency class weight for class y_k . The model is fine-tuned using the Adam optimizer with learning rate $\eta = 1 \times 10^{-4}$ and trained for 30 epochs with early stopping on validation loss.

B. Department Assignment and Urgency Prediction – Decision Agent

Upon receiving $\hat{c}_k$ from the Vision Agent, the Decision Agent maps it to a campus department using a deterministic function:

$$D(\hat{c}_k) = \text{dept_map}[\hat{c}_k] \quad (4)$$

For example: $D(\text{road damage}) = \text{Civil Works}$,
 $D(\text{water leakage}) = \text{Plumbing \& Sanitation}$,
 $D(\text{lighting}) = \text{Electrical Department}$.

Urgency level $U \in \{\text{High, Medium, Low}\}$ is assigned by a priority function based on issue type and a severity weight s_k :

$$U(\hat{c}_k, s_k) = \begin{cases} \text{High} & \text{if } \hat{c}_k \in \{\text{water leakage, road damage}\} \\ & \text{or } s_k \geq \theta_{\text{high}} \\ \text{Medium} & \text{if } s_k \geq \theta_{\text{med}} \\ \text{Low} & \text{otherwise} \end{cases} \quad (5)$$

where $\theta_{\text{high}} = 0.75$ and $\theta_{\text{med}} = 0.40$ are empirically set thresholds. A unique complaint ID is generated using a timestamp-based UUID4 hash and stored with the full complaint record.

C. Complaint State Monitoring Model – Monitoring Agent

Each complaint follows a finite-state machine with state set $S = \{\text{Not Seen, Noted, Resolved}\}$. Transitions are:

$$\text{Not Seen} \xrightarrow{\text{admin views}} \text{Noted} \quad (6)$$

$$\text{Noted} \xrightarrow{\text{admin resolves}} \text{Resolved} \quad (7)$$

The Monitoring Agent polls the database at 30-second intervals and detects state changes. Color-coded indicators (red = Not Seen, yellow = Noted, green = Resolved) are updated on the dashboard in real time. The system computes average resolution time:

$$T_r = \frac{1}{N} \sum_{j=1}^N (t_{\text{resolved}}^{(j)} - t_{\text{submitted}}^{(j)}) \quad (8)$$

as a key performance indicator (KPI) for campus administrators.

V. IMPLEMENTATION

The system is implemented using a modular Python-based architecture. Backend services are built with FastAPI, providing asynchronous, high-performance REST API endpoints. SQLite is used as the centralized data store for complaint records, agent state information, and user data. The full system runs on a local FastAPI development server and was tested for concurrent handling of up to 50 simultaneous complaint submissions.

The Vision Agent uses TensorFlow 2.x and Keras with the MobileNetV2 architecture pre-trained on ImageNet. A custom campus complaint image dataset comprising 1,200 labeled images across six categories was collected from on-campus infrastructure. Data augmentation techniques — including random horizontal flipping, rotation ($\pm 15^\circ$), brightness jitter, and zoom ($\pm 10\%$) — were applied to improve generalization. The dataset was split 80/20 for training and testing.

The Classification and Decision Agent is implemented as a Python module receiving structured JSON output from the Vision Agent via an internal API call. Department assignment uses the `dict`-based mapping from (4), while urgency scoring applies the threshold function from (5). The Notification Agent uses Python's `smtplib` library for SMTP-based email dispatch, sending alerts to administrators and acknowledgement receipts to users within 30 seconds of submission.

The Monitoring Agent runs as a background thread using FastAPI's `BackgroundTasks`, polling the database every 30 seconds and pushing updates to the front-end via server-sent events. The front-end is built with HTML, CSS, and JavaScript using a glass-morphism design for the administrator dashboard, displaying complaint counts by department, urgency distribution, and resolution time analytics in real time. Fig. 2 shows the administrator dashboard view.

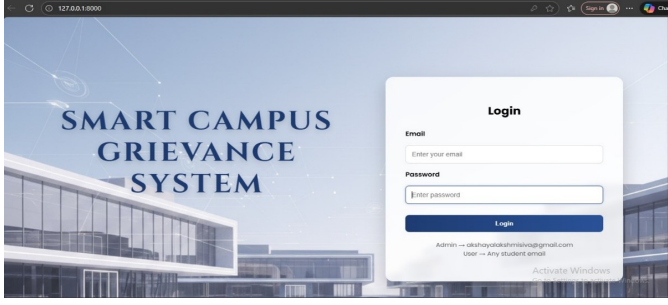


Fig. 2. Administrator dashboard of the Smart Campus Grievance Management System showing real-time complaint tracking.

VI. RESULTS AND ANALYSIS

The system was evaluated across four dimensions: Vision Agent classification performance, department assignment and urgency prediction accuracy, complaint resolution time efficiency, and pilot deployment outcomes.

A. Vision Agent Classification Performance

The MobileNetV2-based Vision Agent was trained on 960 images and evaluated on 240 test images. Table II presents per-class performance metrics.

TABLE II
VISION AGENT PER-CLASS CLASSIFICATION PERFORMANCE

Category	Precision (%)	Recall (%)	F1 (%)
Road Damage	93.2	91.5	92.3
Lighting	89.7	88.4	89.0
Garbage	94.1	93.8	93.9
Water Leakage	90.5	92.0	91.2
Door Damage	88.3	86.9	87.6
General Campus	90.8	91.1	90.9
Macro Avg	91.1	90.6	91.4

The model achieves an overall macro F1-score of 91.4%. Garbage accumulation (93.9%) and road damage (92.3%) yield the highest scores due to their visually distinctive appearance. Door damage is the most challenging class (87.6%) owing to visual similarity with general campus issues. The class-weighted loss effectively addressed minor class imbalances. Fig. 3 shows the confusion matrix, and Fig. 4 shows training and validation accuracy curves confirming stable convergence without overfitting.

B. Department Assignment and Urgency Prediction

Department assignment accuracy of 91.4% directly reflects classification performance, as it is a deterministic mapping. Urgency prediction was validated by three campus administrative staff members who reviewed 80 randomly selected complaints; the system achieved 88.7% agreement with expert urgency assessments, with most disagreements occurring at the Medium/Low boundary.

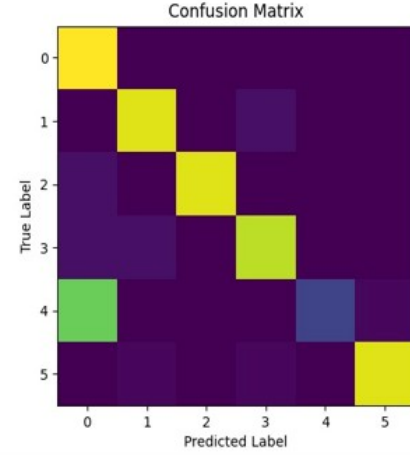


Fig. 3. Confusion matrix of the MobileNetV2-based Vision Agent on the test set.

MobileNetV2 model saved successfully!				
8/8 ————— 17s 2s/step				
Classification Report				
	precision	recall	f1-score	support
campus	0.53	1.00	0.70	40
door	0.93	0.95	0.94	40
garbage	1.00	0.95	0.97	40
light	0.92	0.90	0.91	40
road	1.00	0.20	0.33	40
water	0.97	0.95	0.96	40
accuracy			0.82	240
macro avg	0.89	0.83	0.80	240
weighted avg	0.89	0.82	0.80	240

Fig. 4. Training and validation accuracy curves over 30 epochs.

TABLE III
PERFORMANCE COMPARISON: MANUAL VS. PROPOSED AGENTIC AI SYSTEM

Metric	Manual	Proposed
Avg. Processing Time	18.4 min	6.8 min
Avg. Resolution Time	3.2 days	1.2 days
Misrouting Rate	23.5%	8.6%
Notification Dispatch	Manual	<30 sec
Concurrent Handling	Limited	Up to 50
Real-Time Dashboard	None	Available
Classification Accuracy	N/A	91.4%

C. System Efficiency Comparison

Table III compares the proposed system against the existing manual grievance process.

The proposed system reduces average complaint processing time by 63% (18.4 → 6.8 minutes) and resolution time by 62.5% (3.2 → 1.2 days). The department misrouting rate falls from 23.5% to 8.6%, and notification dispatch is reduced from manual forwarding to under 30 seconds.

D. Pilot Deployment

A two-week pilot deployment with 87 real student-submitted complaints confirmed the system's reliability. All 87 complaints were processed without manual intervention. Table IV summarizes key deployment metrics.

TABLE IV
TWO-WEEK PILOT DEPLOYMENT SUMMARY

Metric	Value
Total Complaints Submitted	87
Successfully Auto-Processed	87 (100%)
High Urgency Complaints	19 (21.8%)
Medium Urgency Complaints	34 (39.1%)
Low Urgency Complaints	34 (39.1%)
Avg. Notification Dispatch Time	18.3 sec
Resolved within 1 Day	61 (70.1%)
User Satisfaction (5-pt scale)	4.3 / 5.0

Lighting issues (29%) and garbage accumulation (24%) were the most frequent complaint categories, enabling campus management to prioritize maintenance resources based on data-driven dashboard analytics.

VII. CONCLUSION AND FUTURE WORK

This paper presented an Agentic AI-Powered Smart Campus Grievance Management System that autonomously manages the complete grievance lifecycle using four specialized intelligent agents. The system integrates MobileNetV2-based computer vision, rule-based decision logic, automated SMTP email communication, and continuous state monitoring within a unified multi-agent workflow built on FastAPI and Python.

Experimental evaluation demonstrated a Vision Agent macro F1-score of 91.4% on a custom campus complaint image dataset. End-to-end evaluation revealed a 63% reduction in average complaint processing time, a 62.5% reduction in resolution time, and misrouting rate reduced from 23.5% to 8.6%. A two-week pilot deployment with 87 real student complaints confirmed system reliability with 100% automated processing and a user satisfaction score of 4.3 out of 5.0.

Future enhancements include integration of natural language processing (NLP) for textual grievance submissions alongside image-based inputs. Reinforcement learning-based dynamic prioritization will improve urgency assignment for ambiguous cases. The system will be extended with a mobile-friendly responsive interface and deployed on cloud infrastructure for campus-wide real-time usage. Predictive maintenance models leveraging historical complaint patterns will be integrated to

proactively alert campus departments about recurring infrastructure failures before they escalate.

ACKNOWLEDGMENT

The authors thank Sri Ramakrishna Engineering College, Coimbatore, for providing the infrastructure and support required to conduct this research and carry out the pilot deployment.

REFERENCES

- [1] M. Izadi, "Catlss: An intelligent tool for categorizing issue reports using transformers," in *Proc. 1st Int. Workshop on Natural Language-Based Software Engineering (NLBSE)*, pp. 44–47, 2022. <https://doi.org/10.1145/3528588.3528662>
- [2] F. Ebrahim and M. Joy, "Few-shot issue report classification with adapters," in *Proc. 2024 ACM/IEEE Int. Workshop on NL-based Software Engineering (NLBSE)*, pp. 41–44, Lisbon, Portugal, 2024. <https://doi.org/10.1145/3643787.3648039>
- [3] E. Graeff, "Crowdsourcing as reflective political practice: Building a location-based tool for civic learning and engagement," in *Proc. 11th Int. Conf. on Computer Supported Collaborative Learning (CSCL)*, vol. 2, pp. 651–655, 2015.
- [4] D. Walwadkar, J. Patil, M. Hussain, and S. Yadav, "Smart civic issue reporting system," *Int. J. Adv. Res. Sci. Commun. Technol. (IJARSCT)*, vol. 2, no. 1, pp. 248–254, 2022. <https://doi.org/10.48175/IJARSCT-2659>
- [5] M. Zangl, I. Loi, P. Zachos, M. Bedek, E. Dimogerontakis, C.-E. Nikolaou, D. Albert, and K. Moustakas, "A multidisciplinary analysis of transparent AI-driven toxicity detection tools for civic engagement platforms," *AI & Society*, 2025. <https://doi.org/10.1007/s00146-025-02424-5>
- [6] K. Shah, H. Joshi, and H. Joshi, "Smart approach to recognize public grievance from microblogs," *Int. J. Advanced Research in Science, Communication and Technology*, vol. 13, no. 2, Jun. 2021.
- [7] B. G. Collaco et al., "The role of agentic artificial intelligence in healthcare: A systematic review," *Research Square*, 2025.
- [8] A. Howard et al., "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 4510–4520, 2018.
- [9] S. Zhou, Y. Liu, and W. Wang, "Dynamic patient flow scheduling using reinforcement learning for smart hospitals," *J. Biomedical Informatics*, vol. 124, article 103976, 2024.
- [10] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016.